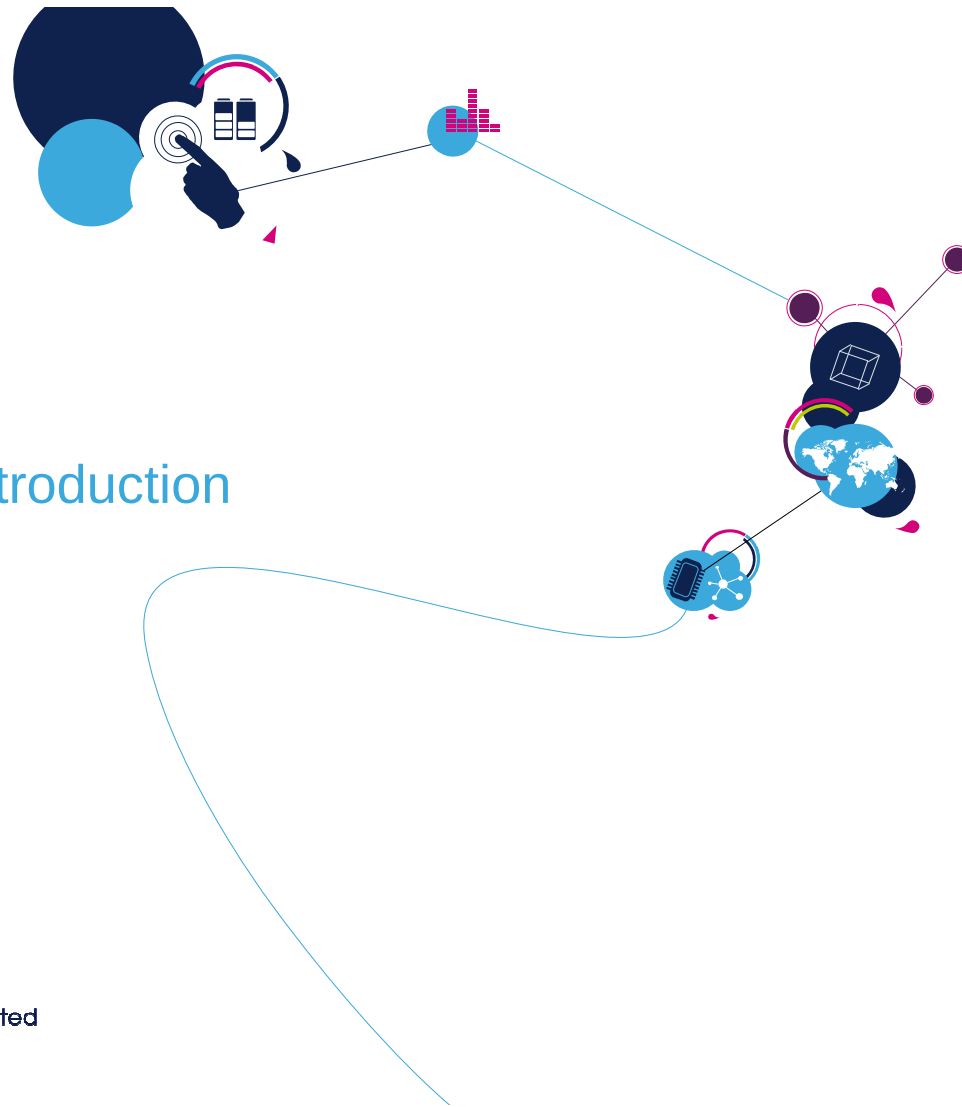
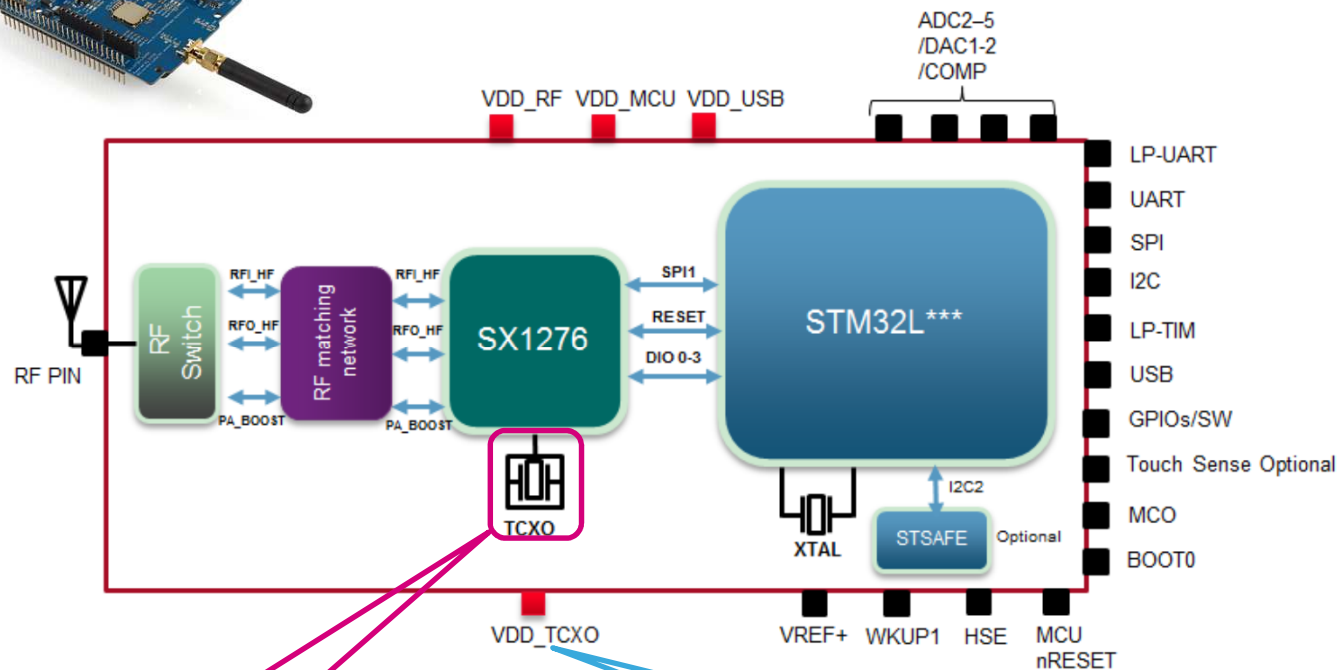


Sigfox introduction



SIGFOX certified Murata module



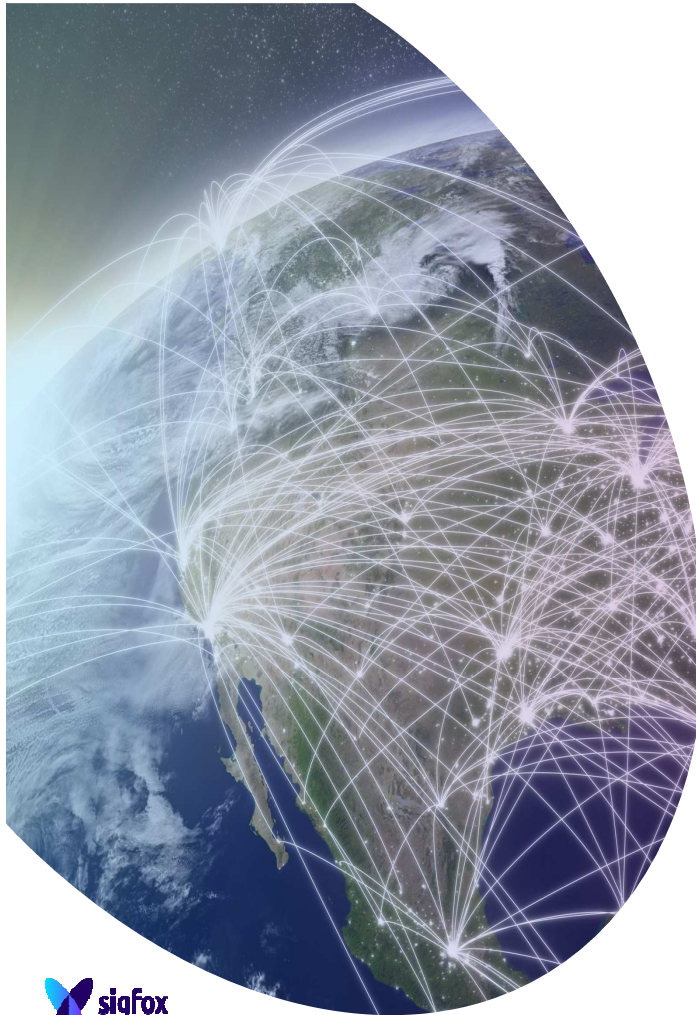
TCXO is active for SIGFOX

Controlled by GPIO



What is
Sigfox?

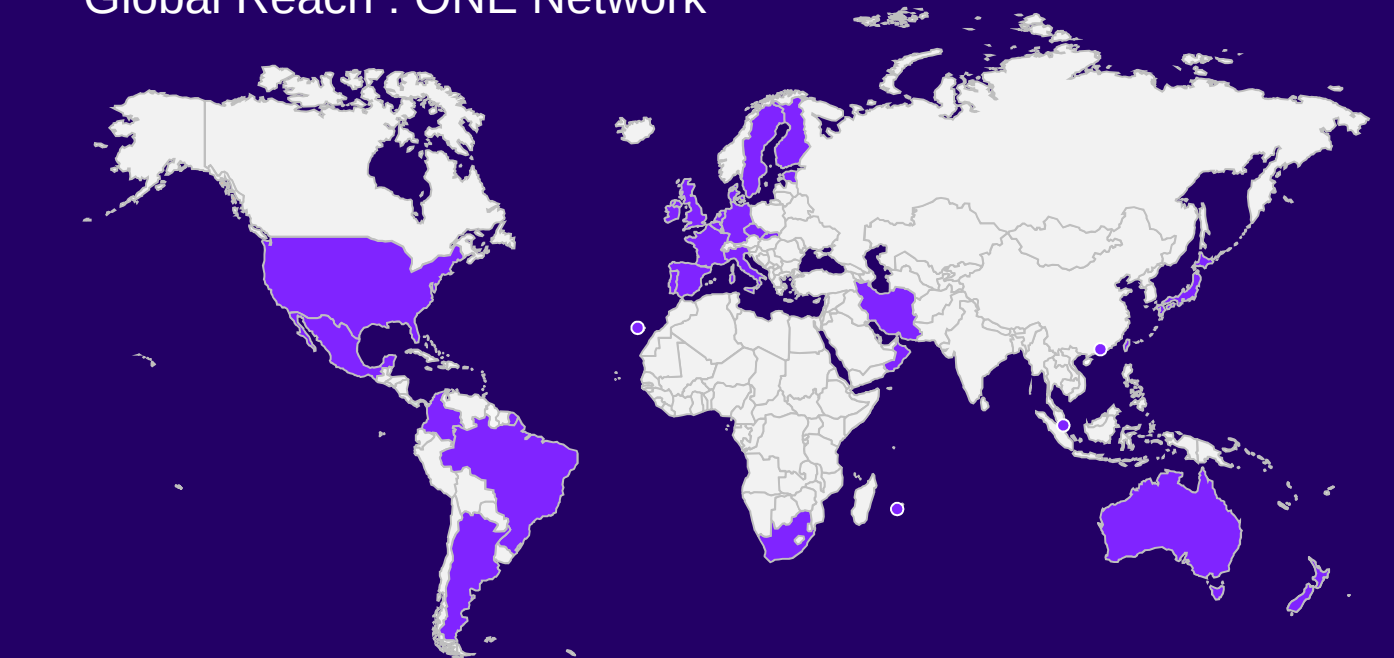




Sigfox: Global LPWA network

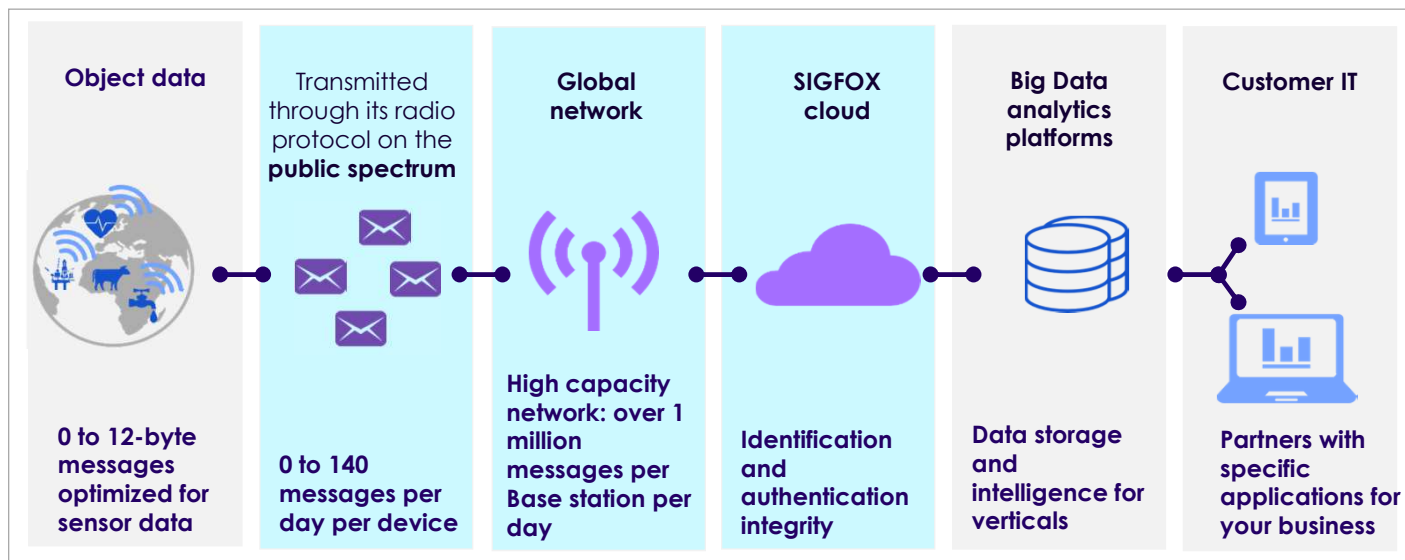
-  **Low power,**
to provide autonomy
-  **Global,**
to be used everywhere
-  **Low cost,**
to address everything
-  **Easy to use,**
and adopted quickly

Global Reach : ONE Network



WHAT DO WE PROVIDE?

A network for connected objects transporting the data from your device to your IT systems

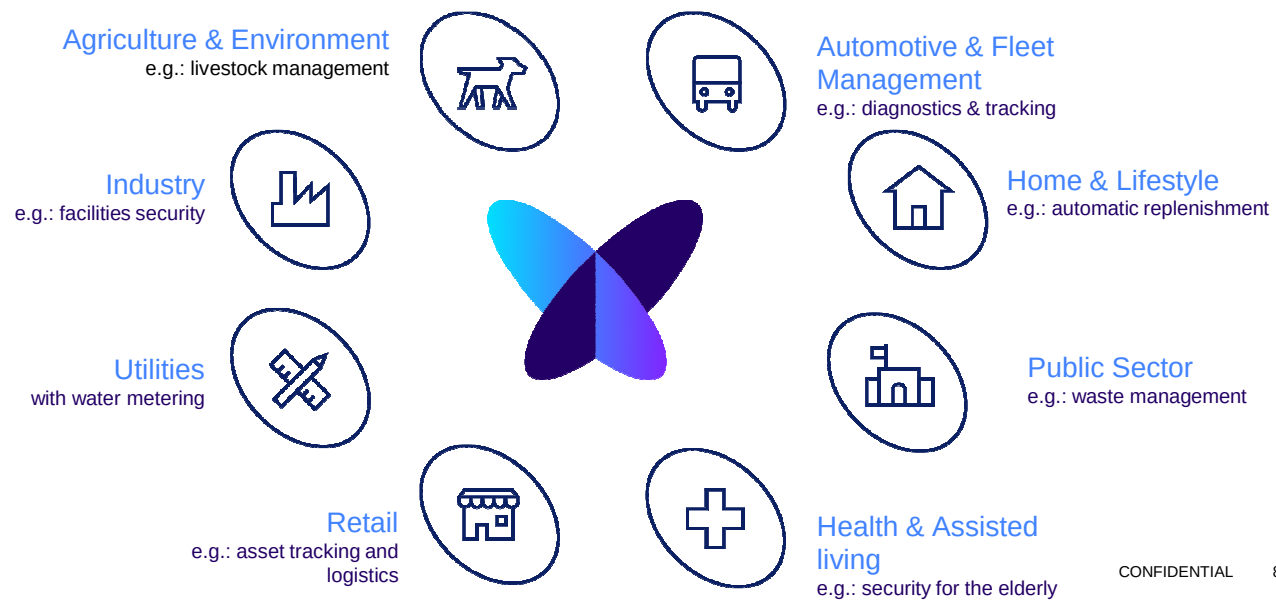


165 devices in the ecosystem and growing...

New additions have been made in every vertical



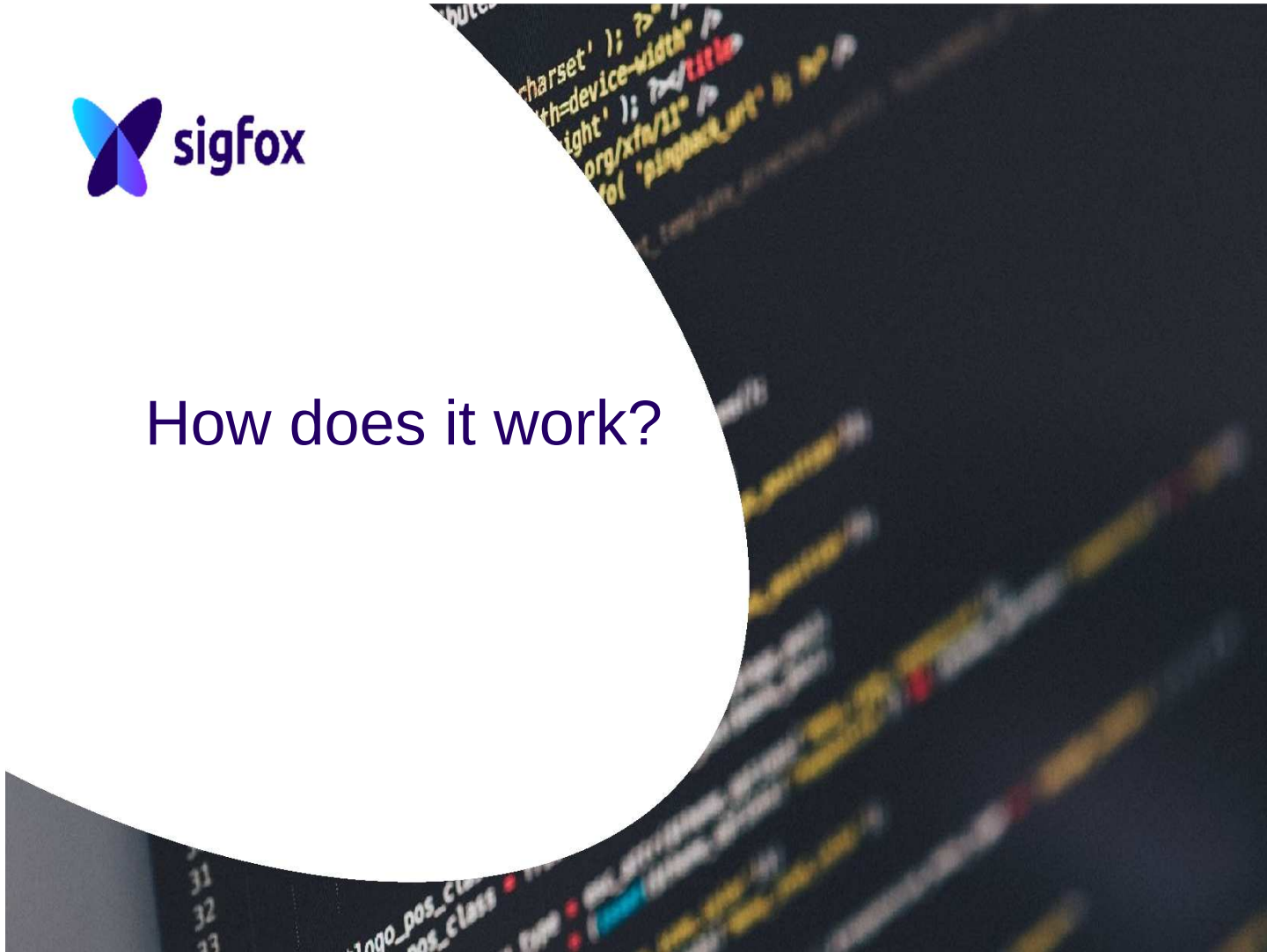
And across all industries...



CONFIDENTIAL 8

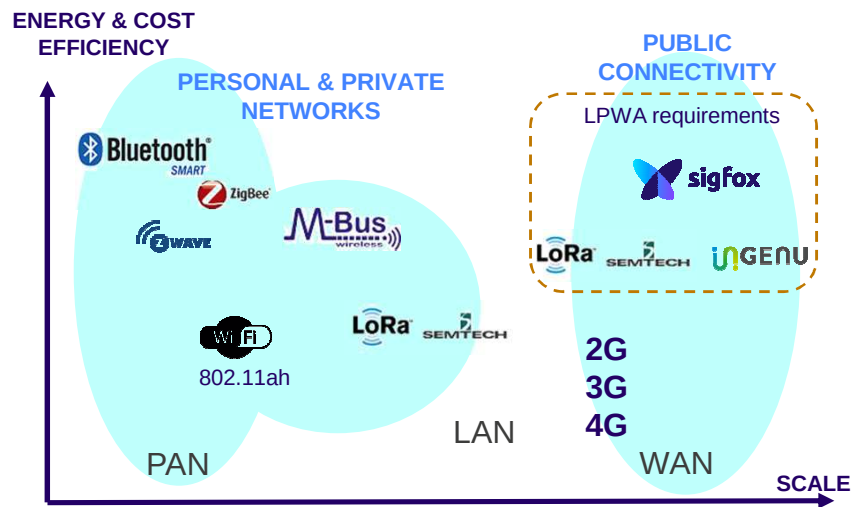


How does it work?



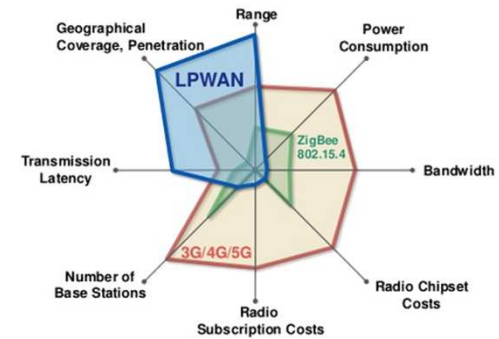
SIGFOX IS A LOW POWER WIDE AREA NETWORK

complementing existing networks to address the bulk of connected objects



LPWA definition

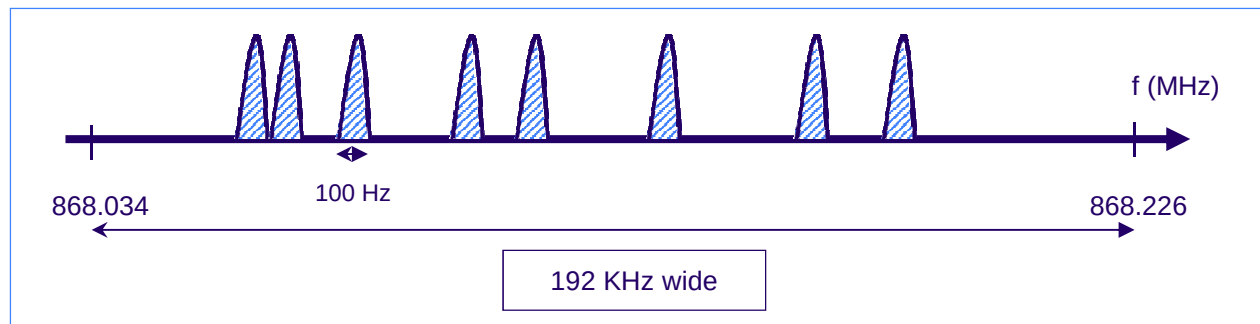
- High density of objects
- Reduced Hardware cost
- Reduced connectivity cost
- Low data rates
- Constrained latency



ULTRA NARROW BAND



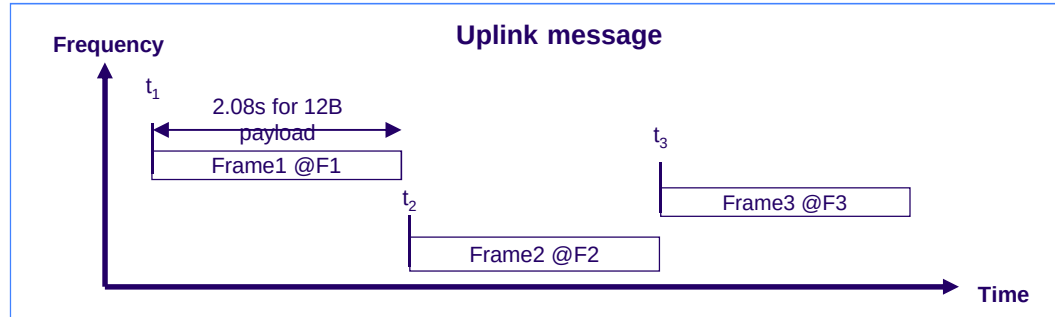
- ✓ Currently spreads on a 200KHz part of the spectrum
- ✓ High spectrum efficiency 1bit/s = 1Hz of bandwidth
- ✓ Each message is ~100Hz wide



RANDOM ACCESS

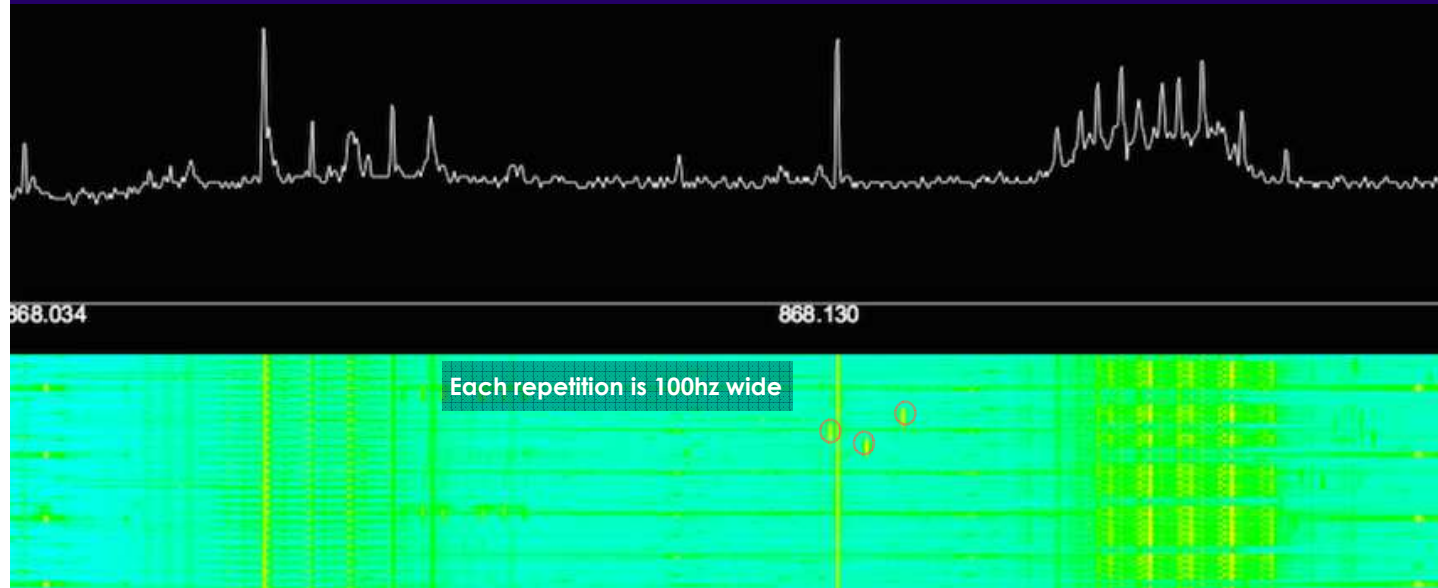


- ✓ Unsynchronized transmission between the network and the device
- ✓ The device transfers a small amount of energy on a random frequency with no protocol overhead (frequency hopping)
- ✓ SIGFOX Base stations permanently listen to the spectrum and interpret received UNB signals
- ✓ The same frame is sent 3 times enabling time and frequency diversity



CONFIDENTIAL 12

Radio spectrum



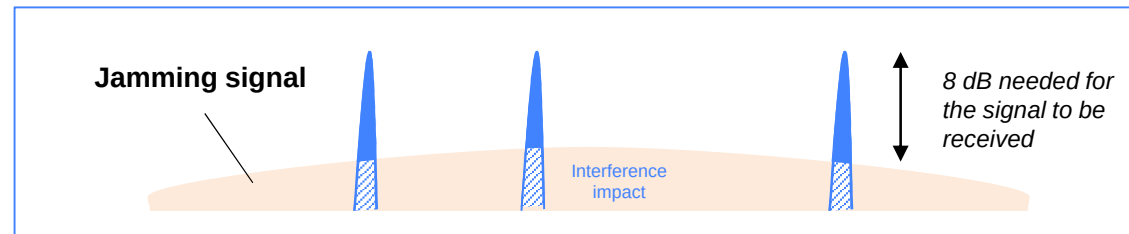


HIGH RESILIENCE TO INTERFERERS

robust to operate in the public ism band



Anti-jamming capabilities due to UNB intrinsic ruggedness coupled with spatial diversity of the base stations (+20dB)

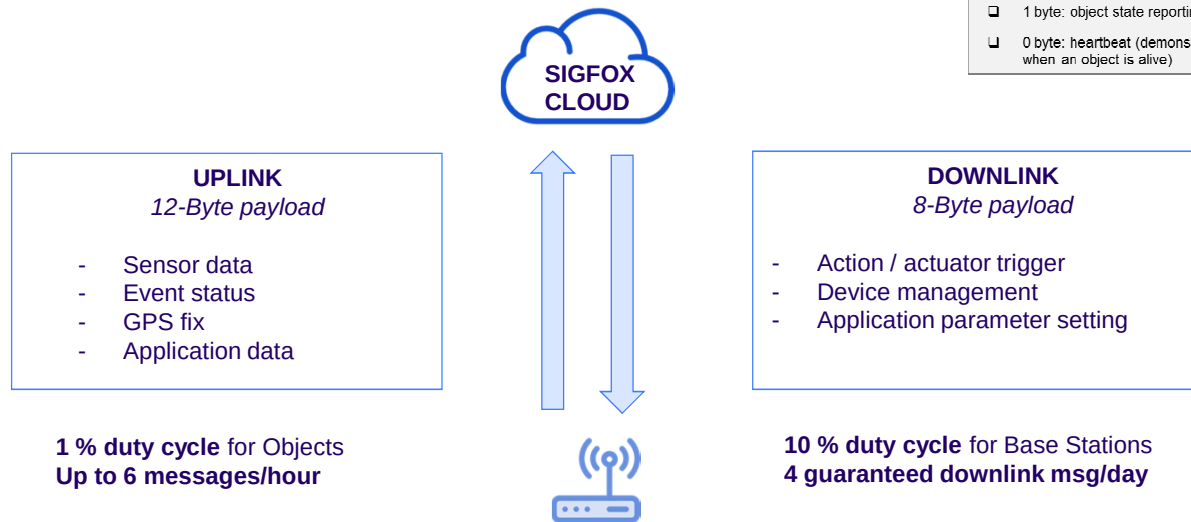


For the same technical reasons as above, UNB is extremely robust in an environment with other spread spectrum signals. However, Spread spectrum networks are affected by UNB signals. **Ultra Narrow Band is therefore the best choice to operate in the public ISM band**



SMALL MESSAGES

to answer the cost & autonomy constraints of remote objects



Payload size examples

- ❑ 6 bytes: GPS coordinates
- ❑ 2 bytes: temperature reporting
- ❑ 1 byte: speed reporting
- ❑ 1 byte: object state reporting
- ❑ 0 byte: heartbeat (demonstrate when an object is alive)

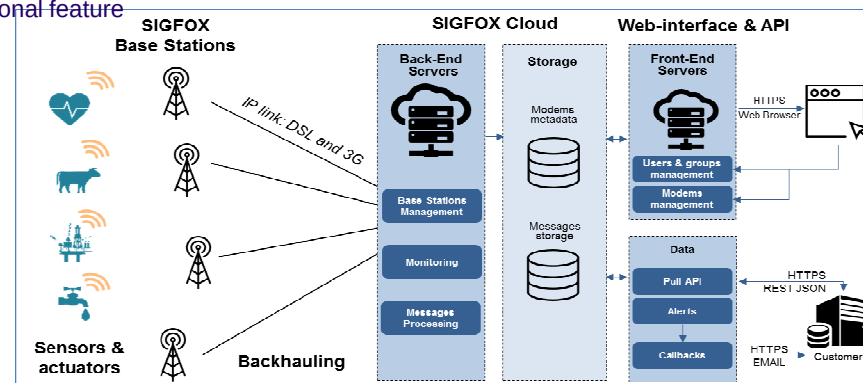
CONFIDENTIAL 15



FLAT ARCHITECTURE

to minimize the cost of deployment and operation

- ✓ SDR cognitive spectrum monitoring overcome the complexity on the device side, generally using expensive TCXOs
- ✓ Data is relayed to the network back-end. Latency is lower than 5 seconds for 90% of the exception reports
- ✓ Base stations also ensure SDR simultaneous multi-carrier transmissions to terminals enabled with the UNB bidirectional feature



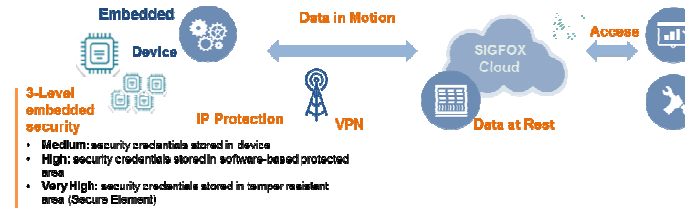
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THE RIGHT LEVEL OF SECURITY

SIGFOX SECURITY IS A PROCESS – **SECURITY BY DEFAULT**
PROTECT WHAT MATTERS, WHERE IT MATTERS



AUTHENTICATION
INTEGRITY
CONFIDENTIALITY
PRIVACY



SIGFOX HAS DESIGNED STRONG SECURITY MECHANISMS AT THE HEART OF THE COMMUNICATION SOLUTION

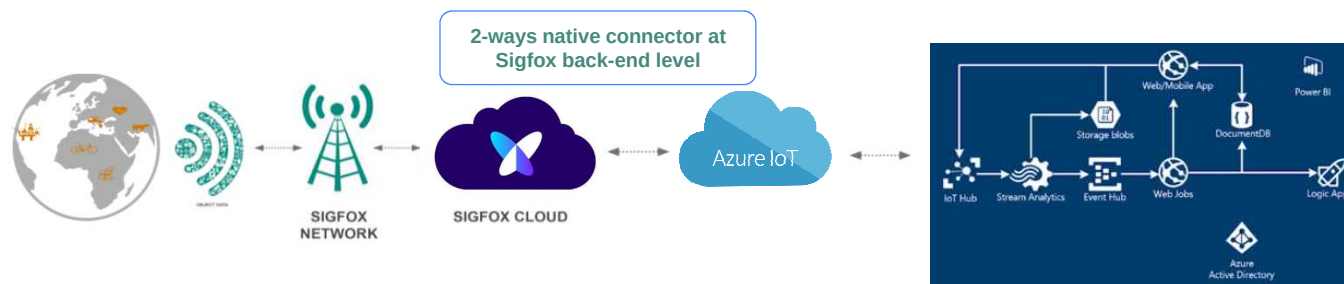


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IOT platforms
partners

Sigfox integrates with Azure IoT hub natively

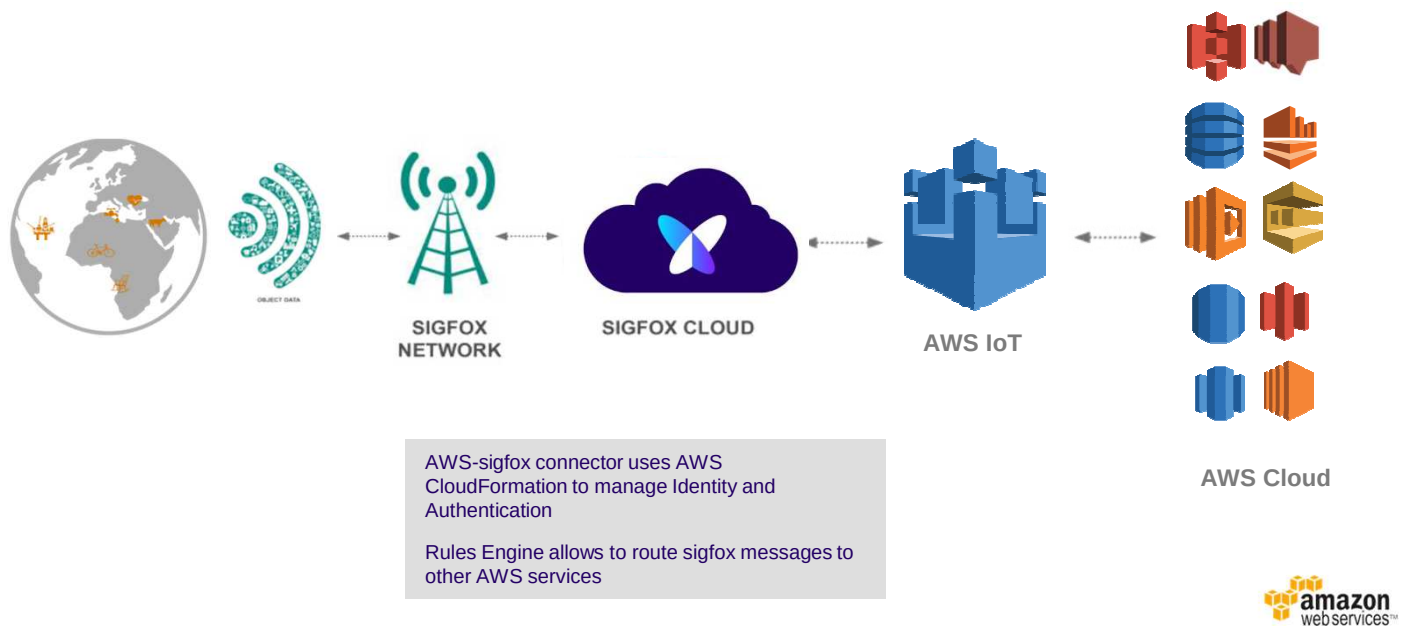


Sigfox <=> Microsoft cloud-to-cloud integration

Each message sent from Sigfox devices having an Azure subscription is automatically forwarded to Microsoft IoT hub



Sigfox integrates with AWS IoT natively



Generic and specialized cloud partners



Full list: <https://partners.sigfox.com/products/platform> (30 certified platforms)

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Hands-on

Thank you





How to get started?



SILICON SOLUTIONS CLUSTERING

Different silicon solutions for different design approaches

Dev Kits / Evaluation Board

- ▶ First steps with sigfox technology
- ▶ Evaluation of Transceiver, SoC, modules



Modules (Sigfox only or Multi-connectivity)



- ▶ Complete modem Sigfox certified and type approved



Transceiver / SoC

- ▶ Standalone chipset(s) used for reference designs, modules and/or combos

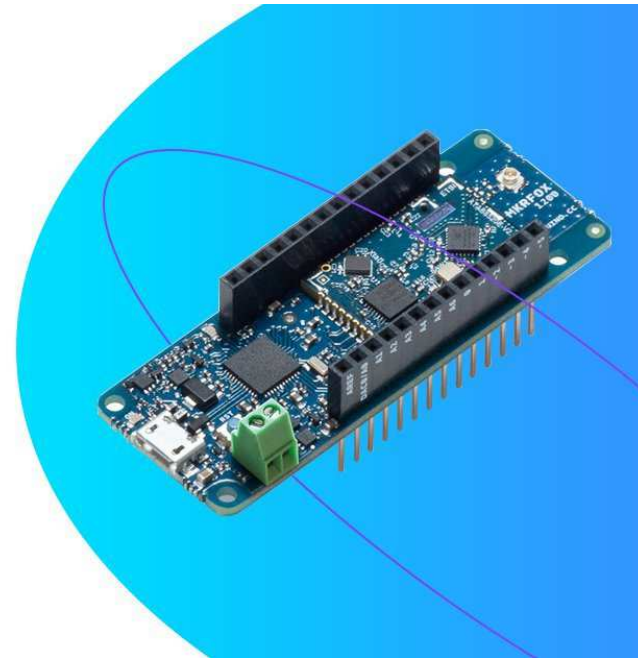




MKRFOX1200

The ideal solution for
makers wanting to
design IoT projects

Available now





API DATA ACCESS

- ✓ SIGFOX API relies on mechanisms widely used on the Internet:
 - HTTP protocol + REST architecture principles
 - HTTPS Authentication per group
 - Returned data in JSON format

- ✓ API allows to access information on the SIGFOX Cloud in an automated programmatic way:
 - Requests are easy to generate and handled by multiple software and languages: C, Python (requests library), curl, browser extensions, ...
 - JSON is easily parse-able: Data structure compatible with all programming languages

THE PARTNER NETWORK

The place to look for Sigfox's ecosystem : <https://partner.sigfox.com>

The image displays two screenshots of the Sigfox Partner Network website. The left screenshot shows the homepage with a network diagram and a 'GET STARTED ON IOT WITH SIGFOX' section. The right screenshot shows a search results page for 'UPLYNX RC22/4 MODULE EVALUATION BOARD'.

SIGFOX PARTNER NETWORK
« Connecting the dots to make the IoT happen »

GET STARTED ON IOT WITH SIGFOX
Discover all Sigfox Ready products and contact the relevant partners for your IoT project.

PRODUCT TYPE

- MODULE (0)
- DEVICE (0)
- ☒ DEV KIT (8)
- SOLUTION (2)

PRODUCT AVAILABILITY

- COMMERCIALLY AVAILABLE (4)
- SAMPLES (3)
- IN DEVELOPMENT (1)

ZONE

- ZONE 1 - EUROPE, CHINA, SOUTH AFRICA (1)
- ☒ ZONE 2 - USA, MEXICO, BRAZIL (8)
- ZONE 4 - AUSTRALIA, NEW ZEALAND, SINGAPORE, TAIWAN, HONG KONG, COLOMBIA

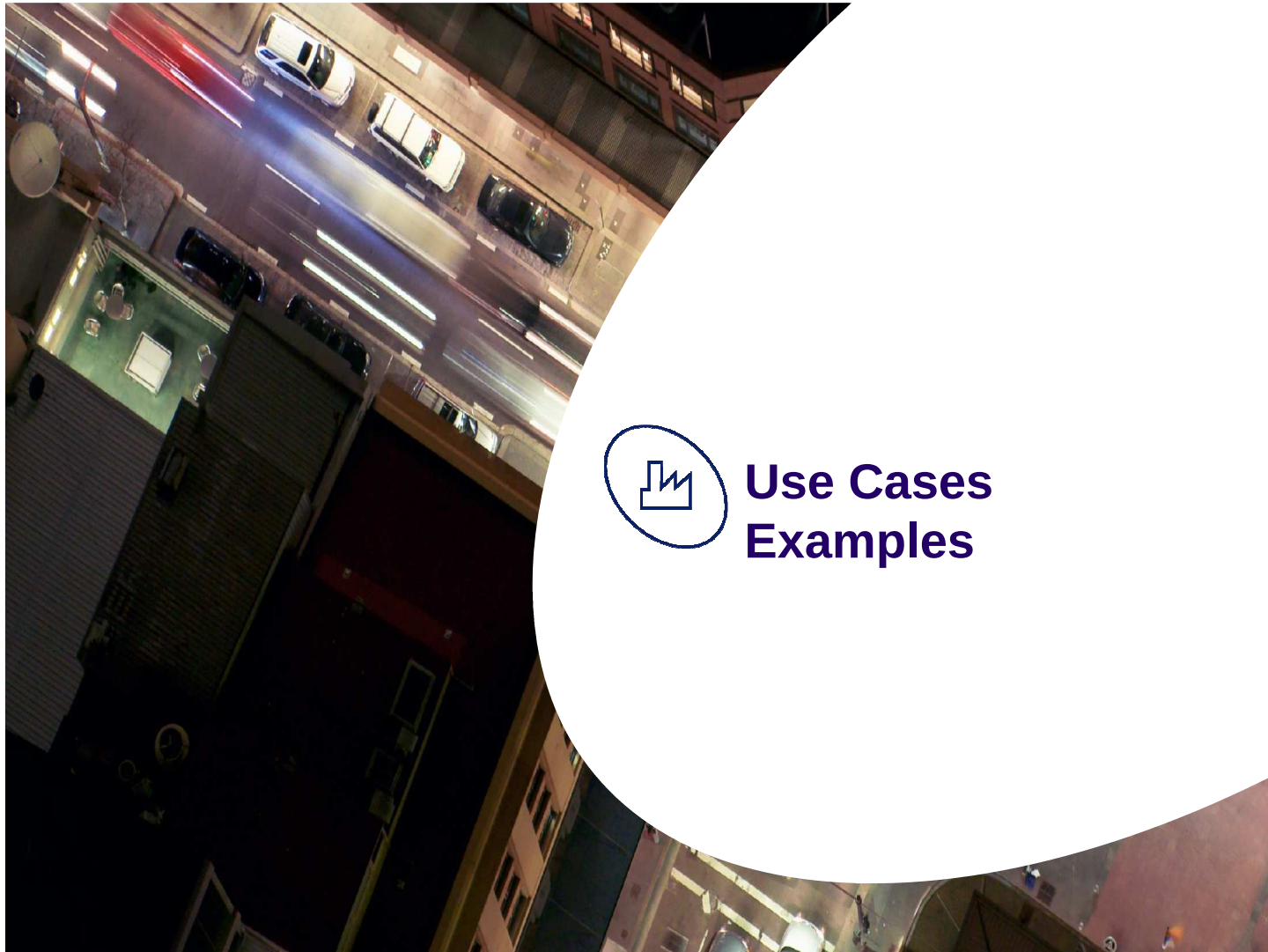
8 products found

Select multiple products to contact at once **START SELECTION**

UPLYNX RC22/4 MODULE EVALUATION BOARD
Uplynx RC22/4 module evaluation board via USB
DEV KIT
SAMPLES

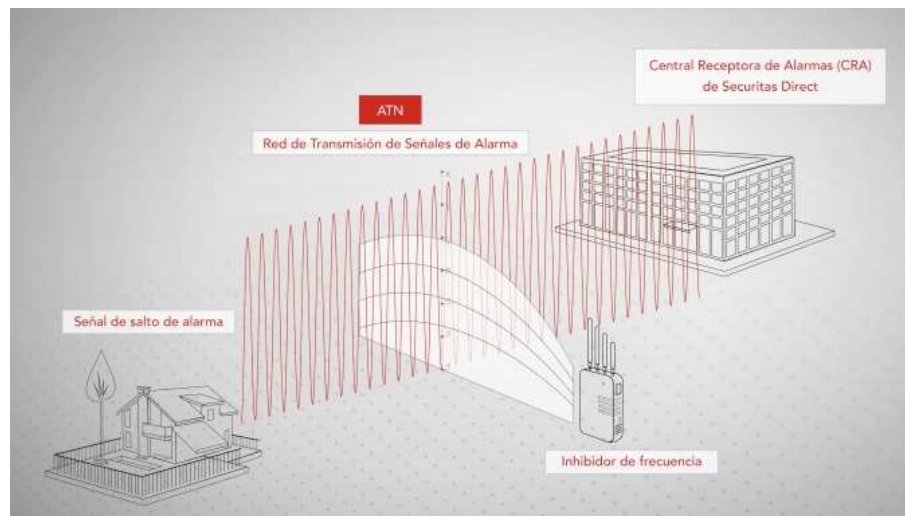
EVBSFM10R2
Sigfox Only Module Evaluation Board for WSSFM10R2(North America)
DEV KIT
SAMPLES

MC-DEMO 205 KIT
Everything you need to get started with mc Demo 205
DEV KIT
SAMPLES



Use Cases Examples

Antijamming solutions



Connected Boilers



Challenge

Create an affordable solution to monitor boilers and create new services for customers.

Solution

Continuous remote monitoring of the boiler's performance and optimize preventive maintenance.

Boilers also communicate and trigger an alert when a failure occurs, allowing professionals to generate responsive, and appropriate remote maintenance.

Benefits

- ✈ Increase customer's satisfaction
- ✈ Control of the energy consumption on a mobile device
- ✈ Money savings
- ✈ Ability to pilot the equipment remotely for the service provider

IMPROVE YOUR UPTIME



*"e.i.m. leblanc early on recognized the value for our professional customers, channel partners and for end users of equipping and installing smart boilers" -
Frédéric Agar, e.i.m. leblanc's president*



Air Quality Monitoring



Challenge

The strong demographic and economic growth of Greater Lyon expected for the next 30 years makes the issue of the measurement of air pollution more and more prevalent. The old system, consisting of fixed sensors, proved expensive and not very representative of rates in different areas of the city, so one had to find a more reliable alternative.

Solution

The new platform Lyon Air relies on mobile sensors on trams and bus, connected through a Sigfox network and responsible for collecting data on the entire metropolis. The data are accessed via the Cloud, citizens can view a dynamic real-time mapping of pollution in the Greater Lyon from a computer or a smartphone.

Benefits

- ✈ An innovative solution to improve the quality of life of citizens
- ✈ A dynamic and interactive mapping of pollution in the city in real time from any device
- ✈ An opportunity to develop new projects for urban monitoring and pollution prediction using Machine Learning

SECURE
BETTER



Connected Defibrillators



Challenge

Defibrillators are often located in remote areas where it is hard to regularly perform auto tests of equipment to ensure they are functioning correctly. Customers who own several defibrillators (e.g. industry) want central supervision. Previously connected boxes were expensive (GSM) and needed to be wired.

Solution

A wall mounted box compatible with Philips HS1 defibrillator, sending monitoring information :

- Door status (open / close)
- Defibrillator's status (OK / NOK)
- Daily Auto test & Battery test

Benefits

- Working defibrillator guaranteed
- Easy installation
 - Added value services: notifications, central supervision
 - Fully wireless: no mains power
- Low power: 4 year autonomy (LR)
 - Plug & Play customer installation

IMPROVE YOUR UPTIME



Alternative partners for this application





Demo



Thank you!

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@aureleq